

Jonathan Zeng

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EDUCATION

Yale University

Expected Graduation Date: May 2028

Bachelor of Science in Computer Science and Mathematics

- **GPA:** 3.93/4.0
- **Relevant Coursework:** Intensive Algorithms, Systems Programming and Computer Organization, Data Structures, Full Stack Web Programming, Building Game Engines, Computational Vision, Analysis 1, Discrete Mathematics, Linear Algebra

EXPERIENCE

Student Researcher

June 2025 – March 2026

Computer Graphics Group, Yale University

- Designing a faster Wavefront .OBJ parser to load and reload 3D objects using multi-threading and SIMD
- Testing the effects of various C++ optimizations on performance, such as thread affinity, profile-guided optimizations, and memory-mapped files, resulting in a 41% performance increase over other parsers
- Communicating results with Professor Mike Shah and other student researchers to write paper for conferences.

Undergraduate Learning Assistant

January 2026 – Present

CPSC3230: Systems Programming and Computer Organization, Yale University

- Tutoring students during weekly office hours, review sessions, and class-wide "hackathon" events
- Providing constructive feedback through grading weekly coding assignments and answering discussion posts
- Maintaining timely communication with professor, teaching fellows, and peer ULAs

Software Engineer

June 2025 – September 2025

Solis Consulting

- Building RoofRoute with Kotlin in Android Studio, an app for roof repair canvassers to find an optimal route of homes to canvass based on parcel and storm data
- Managing backend database on Render.com containing GIS parcel data for over 15 NC counties
- Designing the UI/UX with Jetpack Compose and implementing Google Maps API and voice transcription features

Student Researcher

June 2023 – March 2024

I³T Research Lab, ECE Department, Duke University

- Created and analyzed a game-engine based dataset for monocular Simultaneous Localization and Mapping (SLAM) algorithms using OpenCV for python and the Unreal game engine
- Conducted experiments on various SLAM parameters to find optimal number of features, levels, and scale factor
- Presented poster at NC Science & Engineering Fair, SNCU Research & Creativity Symposium, and lab meetings

PROJECTS

Face Extrusions on Half-edge Meshes | D Programming Language, OpenGL

- 3D graphics app made using OpenGL 4.5 and DLang perform extrusions on half-edge data structures from imported OBJ meshes. Final project for CPSC4792: Real-Time 3D Computer Graphics.
- Used OpenGL and D to implement raycasting system, geometry editor, and camera with custom math library.

Unplugged | Swift

- Built an iOS app with a 5-person team that utilizes the Apple ScreenTime API to lock apps for productivity
- Handled backend logic for users/rooms with Vapor and PostgreSQL to create models, migrations, and controllers

Shader Lab | JavaScript

- Created interactive website showcasing 3D geometry and shaders using Three.js for WebGL integration
- Wrote custom shaders using GLSL and UI layout for shader options with HTML and CSS

TECHNICAL SKILLS

Languages: C, C++, Python, Java, SQL, Kotlin, Lua, D, C#, HTML, CSS

Tools: Git, Linux, Visual Studio, Unity, Unreal, Godot, IntelliJ, Jupyter

Libraries: OpenGL, PyTorch, TensorFlow, SDL, OpenCV, NumPy, Matplotlib, Three.js, Pandas